

Year 7 · Chapter 7 · Decimals: answers (teacher copy)

7.01 Ordering decimals (booklet p. 3)

Examples (model these live)

E1 Which is larger: 0.7 or 0.65?
0.70 and 0.65 · larger: 0.7

E2 Order from smallest to largest: 2.3, 2.03, 2.33, 2.303
2.300, 2.030, 2.330, 2.303 · 2.03, 2.3, 2.303, 2.33

E3 Write < or >: $4.09 \square 4.1$
4.09 and 4.10 · $4.09 < 4.1$

Questions (39 on today's skill, 17 from earlier lessons)

- | | | | |
|--|--|--|--|
| 1 0.79 | 19 9.707, 9.070, 9.700, 9.770 · 9.07, 9.7, 9.707, 9.77 | 32 distances: 0.1, 0.01, 0.1, 0.01 · closest: 2.01 | 43 No. $0.4 = 0.40$, and $40 > 38$. |
| 2 1 tenths (0.1) | 20 0.040, 0.500, 0.940, 0.762 · 0.94, 0.762, 0.5, 0.04 | 33 0.150, 0.495, 0.090, 0.300 · 0.495, 0.3, 0.15, 0.09 | 44 equilateral 6.05 |
| 3 $6.56 > 6.4$ | 21 $x = 135$ 6.07 | 34 $x = 20$ 5.10 | 45 $-0.36, -0.3, 0, 0.2$ |
| 4 0.79 | 22 e.g. 0.45 | 35 e.g. 0.15 | 46 $P = 2 \times 9 + 2 \times 10 \cdot P = 38$ 5.05 |
| 5 1 tenths (0.1) | 23 7.040, 7.404, 7.440, 7.400 · 7.04, 7.4, 7.404, 7.44 | 36 mirror image: each corner the same distance from the line, on the other side 6.01 | 47 $-0.9, -0.39, 0, 0.6$ |
| 6 $6.01 < 6.4$ | 24 $8 \times b \times t$ 5.03 | 37 distances: 0.01, 0.1, 0.01, 0.1 · closest: 2.01 | 48 distances: 0.1, 0.1, 0.01, 0.01 · closest: 2.01 |
| 7 0.76 | 25 0.594, 0.010, 0.100, 0.620 · 0.62, 0.594, 0.1, 0.01 | 38 No. $0.3 = 0.30$, and $30 > 35$. | 49 $a = 100$ 6.09 |
| 8 3 thousandths (0.003) | 26 e.g. 0.25 | 39 19 5.13 | 50 e.g. 0.65 |
| 9 $b - 6$ 5.04 | 27 0.070, 0.170, 0.800, 0.795 · 0.8, 0.795, 0.17, 0.07 | 40 $-0.9, -0.7, 0, 0.3$ | 51 first image on the other side of the line; second image 2 squares below it 6.02 |
| 10 1 thousandths (0.001) | 28 e.g. 0.75 | 41 $= 41 + 59 + 33 + 29 \cdot = 100 + 62 \cdot = 162$ 5.01 | 52 $-0.88, -0.1, 0, 0.5$ |
| 11 $3.29 < 3.5$ | 29 0 6.03 | 42 distances: 0.1, 0.04, 0.05, 0.1 · closest: 5.04 | 53 distances: 0.09, 0.1, 0.01, 0.1 · closest: 0.99 |
| 12 0.18 | 30 9.050, 9.505, 9.550, 9.500 · 9.05, 9.5, 9.505, 9.55 | | 54 a 5.07 |
| 13 $\frac{8}{15}$ 4.12 | 31 $= 4m \times 9y \cdot = 36my \text{ cm}^2$ 5.08 | | 55 e.g. 0.55 |
| 14 2.030, 2.330, 2.300, 2.303 · 2.03, 2.3, 2.303, 2.33 | | | 56 $360^\circ \div 5 \cdot = 72^\circ$ 6.04 |
| 15 4 hundredths (0.04) | | | |
| 16 $6.76 < 6.9$ | | | |
| 17 77.2 4.11 | | | |
| 18 0.7 | | | |

7.02 Decimals and fractions (booklet p. 5)

Examples (model these live)

E1 Write 0.45 as a fraction in simplest form.
 $= \frac{45}{100} \cdot = \frac{9}{20}$

E2 Write $\frac{3}{5}$ as a decimal.
 $= \frac{6}{10} \cdot = 0.6$

E3 Write 2.125 as a mixed numeral.
 $= 2 \frac{125}{1000} \cdot = 2 \frac{1}{8}$

Questions (40 on today's skill, 16 from earlier lessons)

- | | | | |
|---|---|---|---|
| 1 $\frac{1}{10}$ | 18 $\frac{3}{10}$ | 32 $= \frac{375}{1000} \cdot = \frac{3}{8}$ | 44 0.8, 0.7, $0.75 \cdot 0.7, \frac{3}{4}, \frac{4}{5}$ |
| 2 0.28 | 19 $= \frac{65}{100} \cdot = \frac{13}{20}$ | 33 $= \frac{25}{100} \cdot = 0.25$ | 45 $\frac{9}{4}$ 4.01 |
| 3 0.378 | 20 $= \frac{15}{100} \cdot = 0.15$ | 34 $= 19 + 81 + 15 + 79 \cdot = 100 + 94 \cdot = 194$ 5.01 | 46 $= \frac{125}{1000} \cdot = \frac{1}{8}$ |
| 4 $\frac{2}{5}$ | 21 e.g. 0.35 7.01 | 35 $= 8 \frac{46}{100} \cdot = 8 \frac{23}{50}$ | 47 e.g. 0.25 7.01 |
| 5 0.15 | 22 $= 8 \frac{71}{100} \cdot = 8 \frac{71}{100}$ | 36 $\frac{5}{6}$ 4.01 | 48 $0.75, 0.8, 0.7 \cdot 0.7, \frac{3}{4}, \frac{4}{5}$ |
| 6 0.718 | 23 $= \frac{72}{100} \cdot = \frac{18}{25}$ | 37 $= \frac{78}{100} \cdot = \frac{39}{50}$ | 49 $= \frac{625}{1000} \cdot = \frac{5}{8}$ |
| 7 $\frac{1}{5}$ | 24 1 6.03 | 38 $= \frac{875}{1000} \cdot = \frac{7}{8}$ | 50 obtuse-angled · scalene (no equal angles) 6.05 |
| 8 0.14 | 25 $= \frac{75}{100} \cdot = 0.75$ | 39 $0.48, 0.5, 0.6 \cdot 0.48, \frac{1}{2}, \frac{3}{5}$ | 51 $= 6 \frac{36}{100} \cdot = 6 \frac{9}{25}$ |
| 9 9 thousandths (0.009) 7.01 | 26 $= 1 \frac{58}{100} \cdot = 1 \frac{29}{50}$ | 40 $0.900, 0.833, 0.430, 0.020 \cdot 0.9, 0.833, 0.43, 0.02$ 7.01 | 52 $1 \frac{1}{4}$ 4.01 |
| 10 0.99 | 27 $= \frac{40}{100} \cdot = 0.4$ | 41 No. $0.3 = \frac{3}{10}$, but $\frac{1}{3} = 0.333...$ | 53 $0.8, 0.75, 0.7 \cdot 0.7, \frac{3}{4}, \frac{4}{5}$ |
| 11 0.472 | 28 $= 2 \frac{45}{100} \cdot = 2 \frac{9}{20}$ | 42 121 5.05 | 54 $= \frac{64}{1000} \cdot = \frac{8}{125}$ |
| 12 $\frac{7}{10}$ | 29 $1 \frac{2}{3}$ 4.01 | 43 $= \frac{25}{1000} \cdot = \frac{1}{40}$ | 55 2.010, 2.101, 2.100, 2.110 · 2.01, 2.1, 2.101, 2.11 7.01 |
| 13 $m - 9$ 5.04 | 30 $= \frac{2}{100} \cdot = \frac{1}{50}$ | | 56 $= 3 \frac{1}{100} \cdot = 3 \frac{1}{100}$ |
| 14 $= \frac{54}{100} \cdot = \frac{27}{50}$ | 31 7.060, 7.660, 7.606, 7.600 · 7.06, 7.6, 7.606, 7.66 7.01 | | |
| 15 0.48 | | | |
| 16 0.024 | | | |
| 17 $\frac{3}{2}$ 4.01 | | | |

7.03 Adding and subtracting decimals (booklet p. 7)

Examples (model these live)

E1 $3.45 + 2.8$
 $3.45 + 2.80 \cdot = 6.25$

E2 $7 - 2.36$
 $7.00 - 2.36 \cdot = 4.64$

E3 A \$20 note pays for a \$13.85 meal.
 Find the change.
 $20.00 - 13.85 \cdot = \$6.15$

Questions (49 on today's skill, 17 from earlier lessons)

- | | | | |
|---|--|--|--|
| 1 17.4 | 20 $4 \times 27 = 108 \cdot 80 + 28 =$ | 38 0.195 ^{7.02} | 53 $4.70 + 8.58 + 3.00 \cdot = 16.28$ |
| 2 3.9 | 108 ^{5.02} | 39 $2.59 + 1.30 \cdot = 3.89$ | 54 0.94 ^{7.02} |
| 3 15.85 | 21 $3.35 + 1.80 \cdot = 5.15$ | 40 8.060, 8.606, 8.660, 8.600 $\cdot$ | 55 $2.55 + 4.18 = 6.73 \cdot 6.73 - 5$ |
| 4 19.1 | 22 5.3 | 8.06, 8.6, 8.606, 8.66 ^{7.01} | $= 1.73 \text{ km}$ |
| 5 1 | 23 7.27 | 41 $6.00 + 2.30 + 8.00 \cdot = 16.3$ | 56 e.g. 0.15 ^{7.01} |
| 6 13.97 | 24 0.02 ^{7.02} | 42 $4.00 - 2.07 \cdot = 1.93$ | 57 $3.93 + 4.11 = 8.04 \cdot 8.04 -$ |
| 7 6.8 | 25 10 | 43 $= 34 + 66 + 30 + 37 \cdot = 100$ | $5 = 3.04 \text{ km}$ |
| 8 5.3 | 26 $2.53 + 3.30 \cdot = 5.83$ | $+ 67 \cdot = 167$ ^{5.01} | 58 $\square = 10.54 - 2.33 \cdot \square = 8.21$ |
| 9 1 thousandths (0.001) ^{7.01} | 27 2 | 44 $50.00 - 15.16 \cdot = \$34.84$ | 59 $11p = 231 \cdot p = 21$: 21 |
| 10 8.71 | 28 $6.58 + 6.90 \cdot = 13.48$ | 45 $\frac{2}{5}$ ^{7.02} | pencils ^{5.13} |
| 11 4.8 | 29 $14.00 - 2.28 \cdot = 11.72$ | 46 $8.07 + 2.40 \cdot = 10.47$ | 60 line up the decimal points |
| 12 2 | 30 e.g. 0.85 ^{7.01} | 47 $3.20 + 8.15 + 3.00 \cdot = 14.35$ | $\cdot 7.80 + 8.60 = 16.4$ |
| 13 11.25 | 31 $20.00 - 2.07 \cdot = \$17.93$ | 48 $4.56 + 4.97 = 9.53 \cdot 9.53 - 5$ | 61 0.724 ^{7.02} |
| 14 12.1 | 32 $2.82 + 7.30 \cdot = 10.12$ | $= 4.53 \text{ km}$ | 62 $2.90 + 6.12 + 1.00 \cdot = 10.02$ |
| 15 8.5 | 33 $a = 65$ ^{6.06} | 49 0.865, 0.700, 0.860, 0.010 | 63 $50.00 - 17.20 \cdot = \$32.80$ |
| 16 11.45 | 34 $8.00 - 2.88 \cdot = 5.12$ | $\cdot 0.865, 0.86, 0.7, 0.01$ ^{7.01} | 64 7.909, 7.990, 7.900, 7.090 $\cdot$ |
| 17 1.8 | 35 $50.00 - 2.12 \cdot = \$47.88$ | 50 $\square = 17.38 - 8.31 \cdot \square = 9.07$ | 7.09, 7.9, 7.909, 7.99 ^{7.01} |
| 18 8.24 | 36 $10.00 - 2.02 \cdot = 7.98$ | 51 $n = 3$ ^{5.12} | 65 $4.51 + 2.01 = 6.52 \cdot 6.52 - 5$ |
| 19 5.6 | 37 $20.00 - 3.06 \cdot = \$16.94$ | 52 line up the decimal points | $= 1.52 \text{ km}$ |
| | | $\cdot 7.30 + 5.88 = 13.18$ | 66 $b - 6$ ^{5.04} |

7.04 Multiplying and dividing by powers of 10 (booklet p. 9)

Examples (model these live)

E1 3.47×100
 digits 2 places left $\cdot = 347$

E2 $52.6 \div 1000$
 digits 3 places right $\cdot = 0.0526$

E3 Change 4.5 kg to grams.
 $1 \text{ kg} = 1000 \text{ g} \cdot 4.5 \times 1000 = 4\,500 \text{ g}$

Questions (49 on today's skill, 17 from earlier lessons)

- | | | | |
|----------------------------------|--|---|---|
| 1 6 910 | 21 1 km = 1 000 m $\cdot 9.7 \times 1$ | 36 10 mm = 1 cm $\cdot 210 \div 10 =$ | 52 $\square = 79 \div 1\,000 \cdot \square = 0.079$ |
| 2 9.64 | 000 = 9 700 m | 21 cm | 53 $= 3\,780 \div 1000 \cdot = 3.78$ |
| 3 0.14 | 22 0.0901 | 37 1 000 | 54 $8.24 < 8.7$ ^{7.01} |
| 4 89.2 | 23 0.28 | 38 7 thousandths (0.007) ^{7.01} | 55 1000 pens: \$5 370 $\cdot 10$ |
| 5 0.0702 | 24 680 ^{3.02} | 39 1 cm = 10 mm $\cdot 3.08 \times 10 =$ | pens: \$53.70 |
| 6 0.6 | 25 874 | 30.8 mm | 56 $1.18 + 7.30 \cdot = 8.48$ ^{7.03} |
| 7 48.8 | 26 1 km = 1 000 m $\cdot 0.34 \times 1$ | 40 $20.00 - 2.06 \cdot = \$17.94$ ^{7.03} | 57 $= 9\,440 \div 1000 \cdot = 9.44$ |
| 8 0.23 | 000 = 340 m | 41 $\square = 95 \div 1\,000 \cdot \square = 0.095$ | 58 1000 pens: \$8 110 $\cdot 10$ |
| 9 5.1 ^{7.03} | 27 0.0605 | 42 10 mm = 1 cm $\cdot 254 \div 10 =$ | pens: \$81.10 |
| 10 2.1 | 28 1 kg = 1 000 g $\cdot 8.97 \times 1$ | 25.4 cm | 59 $= 5c + 4t + 5c + 4t \cdot = 10c +$ |
| 11 573 | 000 = 8 970 g | 43 $18 \div 6 = 3 \cdot = 3x$ ^{5.09} | $8t$ ^{5.07} |
| 12 0.0898 | 29 100 cm = 1 m $\cdot 32 \div 100 =$ | 44 10 | 60 $\square = 89 \div 1\,000 \cdot \square = 0.089$ |
| 13 0.15 | 0.32 m | 45 560 ^{3.02} | 61 $= 40 \times 6 + 4 \times 6 \cdot = 240 +$ |
| 14 13.1 | 30 $= \frac{62}{100} \cdot = 0.62$ ^{7.02} | 46 1 cm = 10 mm $\cdot 8.75 \times 10 =$ | $24 \cdot = 264$ ^{3.02} |
| 15 4.2 | 31 100 | 87.5 mm | 62 10 |
| 16 0.37 | 32 1 L = 1 000 mL $\cdot 2.83 \times 1$ | 47 $\square = 3.3 \div 100 \cdot \square = 0.033$ | 63 $= 6\,980 \div 1000 \cdot = 6.98$ |
| 17 6.61 | 000 = 2 830 mL | 48 $= 4\,250 \div 1000 \cdot = 4.25$ | 64 $= \frac{56}{100} \cdot = \frac{14}{25}$ ^{7.02} |
| 18 0.21 | 33 $a = 120$ ^{6.09} | 49 $= 4 \cdot \frac{26}{100} \cdot = 4 \cdot \frac{13}{50}$ ^{7.02} | 65 1000 pens: \$6 840 $\cdot 10$ |
| 19 76.8 | 34 10 mm = 1 cm $\cdot 630 \div 10 =$ | 50 1000 pens: \$1 500 $\cdot 10$ | pens: \$68.40 |
| 20 4 right, 3 up ^{6.01} | 63 cm | 51 $m - 7$ ^{5.04} | 66 a full turn: back where it |
| | 35 100 | | started ^{6.02} |

7.05 Using a known product (booklet p. 11)

Examples (model these live)

E1 Given $36 \times 7 = 252$, find 3.6×7 .
 1 decimal place $\cdot = 25.2$

E2 Given $36 \times 7 = 252$, find 0.36×0.7 .
 2 + 1 = 3 decimal places $\cdot = 0.252$

E3 Given $252 \div 7 = 36$, find $25.2 \div 7$.
 25.2 is $252 \div 10 \cdot = 3.6$

Questions (40 on today's skill, 17 from earlier lessons)

- | | | | |
|--------|--------|-------------------------|---------------------------------------|
| 1 27.9 | 5 10.5 | 9 0.176 ^{7.04} | 13 equilateral ^{6.05} |
| 2 13.5 | 6 1.2 | 10 7.5 | 14 $1 + 1 = 2$ decimal places $\cdot$ |
| 3 0.63 | 7 13.5 | 11 2.4 | $= 1.11$ |
| 4 23.4 | 8 24.6 | 12 31.2 | 15 13.2 |

- 16 3.42
 17 2.3 ^{7.03}
 18 15.5
 19 $1 + 1 = 2$ decimal places $\cdot = 0.98$
 20 $21.5 = 215 \div 10 \cdot = 4.3$
 21 $0.391, 0.850, 0.900, 0.050 \cdot 0.9, 0.85, 0.391, 0.05$ ^{7.01}
 22 2 decimal places $\cdot = 18.13$
 23 $1 + 1 = 2$ decimal places $\cdot = 2.88$
 24 $= \frac{12}{100} \cdot = 0.12$ ^{7.02}
 25 $28.8 = 288 \div 10 \cdot = 3.6$
 26 2 decimal places $\cdot = 16.25$
 27 $10.2 = 102 \div 10 \cdot = 3.4$
 28 2 decimal places $\cdot = 22.2$
 29 49 ^{5.05}
 30 $1 + 1 = 2$ decimal places $\cdot = 1.68$
 31 $6.1 < 6.3$ ^{7.01}
 32 $1 + 2 = 3$ decimal places $\cdot = 1.014$
 33 $25.8 = 258 \div 10 \cdot = 4.3$
 34 4.8 ^{7.04}
 35 2 decimal places $\cdot = 5.19$
 36 $50.00 - 1.56 \cdot = \$48.44$ ^{7.03}
 37 $1 + 1 = 2$ decimal places $\cdot = 0.6$
 38 $1 + 2 = 3$ decimal places $\cdot = 0.486$
 39 $\times 10$ and $\div 10$ cancel $\cdot = 156$
 40 first: $4t \cdot = 4t + 11$ ^{5.04}
 41 $= 8 \div 5 \cdot = 1.6$
 42 0.428 ^{7.02}
 43 $1 + 2 = 3$ decimal places $\cdot = 0.85$
 44 $\times 10$ and $\div 10$ cancel $\cdot = 437$
 45 $2.900, 2.990, 2.090, 2.909 \cdot 2.09, 2.9, 2.909, 2.99$ ^{7.01}
 46 $= 24 \div 6 \cdot = 4$
 47 $1 \text{ m} = 100 \text{ cm} \cdot 5.53 \times 100 = 553 \text{ cm}$ ^{7.04}
 48 $\times 10$ and $\div 10$ cancel $\cdot = 550$
 49 $= 8 \div 4 \cdot = 2$
 50 n^2 ^{5.03}
 51 $1 + 2 = 3$ decimal places $\cdot = 0.336$
 52 3 hundredths (0.03) ^{7.01}
 53 2 decimal places $\cdot = 35.2$
 54 $\times 10$ and $\div 10$ cancel $\cdot = 368$
 55 16.8 ^{7.03}
 56 $= 12.4 \div 4 \cdot = 3.1$
 57 $= \frac{34}{100} \cdot = \frac{17}{50}$ ^{7.02}

7.06 Multiplying decimals (booklet p. 13)

Examples (model these live)

- E1 0.3×0.4
 $3 \times 4 = 12$, 2 decimal places $\cdot = 0.12$
 E2 2.5×1.2
 $25 \times 12 = 300$, 2 decimal places $\cdot = 3.00 = 3$
 E3 4.6×0.3
 $46 \times 3 = 138$, 2 decimal places $\cdot = 1.38$

Questions (49 on today's skill, 17 from earlier lessons)

- 1 0.5
 2 0.4
 3 38.4
 4 5.4
 5 0.12
 6 12.2
 7 3.6
 8 0.03
 9 9.3 ^{7.05}
 10 7.5
 11 0.9
 12 0.35
 13 9.6
 14 6.4
 15 0.45
 16 62.4
 17 0.06
 18 23.1
 19 4.9
 20 62.1 ^{4.11}
 21 $26 \times 3 = 78$, 2 decimal places $\cdot = 0.78$
 22 0.81
 23 20.7
 24 3.39 ^{7.04}
 25 2.5
 26 $71 \times 5 = 355$, 2 decimal places $\cdot = 3.55$
 27 0.16
 28 $31 \times 7 = 217$, 2 decimal places $\cdot = 2.17$
 29 $793 \times 4 = 3172$, 2 decimal places $\cdot = 31.72$
 30 $0.494, 0.500, 0.210, 0.040 \cdot 0.5, 0.494, 0.21, 0.04$ ^{7.01}
 31 $11 \times 18 = 198 \cdot = 1.98 \text{ m}^2$
 32 $0.8 \times 0.8 \cdot = 0.64$
 33 $17.00 - 1.38 \cdot = 15.62$ ^{7.03}
 34 $92 \times 4 = 368$, 2 decimal places $\cdot = 3.68$
 35 $456 \times 4 = 1824$, 2 decimal places $\cdot = 18.24$
 36 $148 \times 9 = 1332$, 2 decimal places $\cdot = 13.32$
 37 $47 \times 15 = 705 \cdot = 7.05 \text{ m}^2$
 38 $a = 75$ ^{6.06}
 39 $0.4 \times 0.4 \cdot = 0.16$
 40 0.61 ^{7.02}
 41 $40 \times 9 = 360$, 2 decimal places $\cdot = 3.6$
 42 $4.8 \times 5.70 = 27.36 \cdot \approx \27.36
 43 2.52 ^{7.05}
 44 $552 \times 3 = 1656$, 2 decimal places $\cdot = 16.56$
 45 10 ^{7.04}
 46 $24 \times 11 = 264 \cdot = 2.64 \text{ m}^2$
 47 $1.4 \times 2.09 = 2.926 \cdot \approx \2.93
 48 $75 \times 73 = 5475$, 3 decimal places $\cdot = 5.475$
 49 $360^\circ \div 3 \cdot = 120^\circ$ ^{6.04}
 50 Smaller: multiplying by 0.9, which is less than 1, makes it smaller (0.72).
 51 10.55 ^{7.03}
 52 $3.6 \times 8.07 = 29.052 \cdot \approx \29.05
 53 $12 \times 14 = 168$, 3 decimal places $\cdot = 0.168$
 54 e.g. 0.15 ^{7.01}
 55 Smaller: multiplying by 0.4, which is less than 1, makes it smaller (1).
 56 $1 + 1 = 2$ decimal places $\cdot = 1.7$ ^{7.05}
 57 $16 \times 45 = 720$, 3 decimal places $\cdot = 0.72$
 58 Smaller: multiplying by 0.3, which is less than 1, makes it smaller (0.18).
 59 40 cm ^{6.10}
 60 $3.3 \times 5.72 = 18.876 \cdot \approx \18.88
 61 0.876 ^{7.02}
 62 $0.7 \times 0.7 \cdot = 0.49$
 63 $29 \times 27 = 783$, 3 decimal places $\cdot = 0.783$
 64 328 ^{7.04}
 65 Smaller: multiplying by 0.5, which is less than 1, makes it smaller (0.2).
 66 $2.99 + 3.00 \cdot = 5.99$ ^{7.03}

7.07 Dividing decimals by whole numbers (booklet p. 15)

Examples (model these live)

- E1 $8.4 \div 4$
 $8 \div 4 = 2$, 4 tenths $\div 4 = 1$ tenth $\cdot = 2.1$
 E2 $7.5 \div 6$
 $7.50 \div 6 \cdot = 1.25$
 E3 $\$14.70$ is shared by 3 people. How much each?
 $14.70 \div 3 \cdot = \$4.90$

Questions (49 on today's skill, 17 from earlier lessons)

- 1 1.8
 2 6.84
 3 0.7
 4 4.5
 5 2.47
 6 0.2
 7 6.7
 8 8.51
 9 0.56 ^{7.06}
 10 0.8
 11 6.9
 12 8.04
 13 0.3
 14 7.6
 15 1.2
 16 0.4
 17 6.84
 18 0.1
 19 9.3
 20 square ^{6.08}
 21 $7.400 \div 5 \cdot = 1.48$
 22 5.19
 23 0.3
 24 5 ^{3.03}
 25 6.2
 26 $8.900 \div 4 \cdot = 2.225$
 27 6.19
 28 $3.000 \div 4 \cdot = 0.75$
 29 $42.30 \div 5 \cdot = \$8.46$
 30 $0.050, 0.800, 0.240, 0.861 \cdot 0.861, 0.8, 0.24, 0.05$ ^{7.01}
 31 divide each place in turn $\cdot = 3.16$
 32 $6.900 \div 4 \cdot = 1.725$
 33 $14.1 = 141 \div 10 \cdot = 4.7$ ^{7.05}
 34 $12.52 \div 2 \cdot = \$6.26$
 35 divide each place in turn $\cdot = 1.769$
 36 $44.04 \div 6 \cdot = \$7.34$

- 37 divide each place in turn $\cdot$ = 4.474
- 38 $6 \times 39 = 234 \cdot 180 + 54 =$
234 ^{5.02}
- 39 $6.100 \div 4 \cdot = 1.525$
- 40 0.56 ^{7.02}
- 41 total: $31.6 \cdot 31.6 \div 4 = 7.9$
- 42 $20.22 \div 6 \cdot = \$3.37$
- 43 2.32 ^{7.04}
- 44 divide each place in turn $\cdot$ = 3.106
- 45 $49 \times 14 = 686 \cdot = 6.86 \text{ m}^2$
7.06
- 46 $2.600 \div 8 \cdot = 0.325$
- 47 total: $29 \cdot 29 \div 4 = 7.25$
- 48 $4.4 \div 8 \cdot = 0.55 \text{ m}$
- 49 $y + 70 = 125 \cdot y = 55^\circ$ ^{6.07}
- 50 $\square = 6.88 \div 8 \cdot \square = 0.86$
- 51 5 ^{3.03}
- 52 total: $13.4 \cdot 13.4 \div 4 = 3.35$
- 53 $3.4 \div 5 \cdot = 0.68 \text{ m}$
- 54 $10.00 - 2.48 \cdot = 7.52$ ^{7.03}
- 55 $\square = 6.3 \div 9 \cdot \square = 0.7$
- 56 2 decimal places $\cdot = 14.49$
7.05
- 57 $1.7 \div 4 \cdot = 0.425 \text{ m}$
- 58 $\square = 7.92 \div 9 \cdot \square = 0.88$
- 59 23.2 ^{4.11}
- 60 total: $26.6 \cdot 26.6 \div 4 = 6.65$
- 61 $4.24 < 4.6$ ^{7.01}
- 62 divide each place in turn $\cdot$ = 4.436
- 63 $2.9 \div 5 \cdot = 0.58 \text{ m}$
- 64 0.03 ^{7.04}
- 65 $\square = 1.71 \div 9 \cdot \square = 0.19$
- 66 $0.8 \times 0.8 \cdot = 0.64$ ^{7.06}

7.08 Dividing by a decimal (booklet p. 17)

Examples (model these live)

E1 $4.8 \div 0.6$

$\times 10: 48 \div 6 \cdot = 8$

E2 $1.25 \div 0.05$

$\times 100: 125 \div 5 \cdot = 25$

E3 How many 0.3 L cups can be filled from 2.4 L?

$2.4 \div 0.3 = 24 \div 3 \cdot = 8 \text{ cups}$

Questions (48 on today's skill, 17 from earlier lessons)

- | | | | |
|------------------------|--|--|--|
| 1 80 | 20 9 ^{5.05} | 39 $\times 10: 72 \div 3 \cdot = 24$ | 53 $16.00 - 1.13 \cdot = 14.87$ ^{7.03} |
| 2 4 | 21 $\times 10: 234 \div 9 \cdot = 26$ | 40 0.92 ^{7.02} | 54 $1.80 \div 0.15 = 180 \div 15 \cdot = 12 \text{ pencils}$ |
| 3 70 | 22 7 | 41 $\times 10: 84 \div 14 \cdot = 6$ | 55 $3.600 \div 4 \cdot = 0.9$ ^{7.07} |
| 4 14 | 23 79 | 42 $\times 100: 54 \div 6 \cdot = 9$ | 56 Larger: dividing by a number less than 1 gives more (40). |
| 5 6 | 24 0.21 ^{7.06} | 43 0.7 ^{7.07} | 57 $0.88 \div 0.08 = 88 \div 8 \cdot = 11 \text{ pencils}$ |
| 6 30 | 25 20 | 44 $3.5 \div 0.5 \cdot = 7 \text{ cups}$ | 58 2 ^{6.04} |
| 7 50 | 26 $\times 10: 184 \div 8 \cdot = 23$ | 45 $31 \times 34 = 1054 \cdot = 10.54 \text{ m}^2$ ^{7.06} | 59 $\times 10: 48 \div 16 \cdot = 3$ |
| 8 8 | 27 9 | 46 $\times 10: 130 \div 13 \cdot = 10$ | 60 3.3 ^{7.04} |
| 9 4.37 ^{7.07} | 28 $\times 10: 104 \div 8 \cdot = 13$ | 47 Larger: dividing by a number less than 1 gives more (30). | 61 $3.2 \div 0.4 \cdot = 8 \text{ cups}$ |
| 10 15 | 29 $\times 100: 114 \div 6 \cdot = 19$ | 48 $\frac{x}{6} = 5 \cdot x = 5 \times 6 = 30$ ^{5.12} | 62 Larger: dividing by a number less than 1 gives more (10). |
| 11 90 | 30 0.838, 0.600, 0.430, 0.060 $\cdot$ 0.838, 0.6, 0.43, 0.06 ^{7.01} | 49 $1.40 \div 0.07 = 140 \div 7 \cdot = 20 \text{ pencils}$ | 63 3.2 ^{7.06} |
| 12 2 | 31 $2.5 \div 0.5 \cdot = 5 \text{ cups}$ | 50 0.57 ^{7.05} | 64 $0.75 \div 0.15 = 75 \div 15 \cdot = 5 \text{ pencils}$ |
| 13 74 | 32 $\times 10: 120 \div 5 \cdot = 24$ | 51 $\times 10: 275 \div 25 \cdot = 11$ | 65 $1 + 1 = 2 \text{ decimal places} \cdot = 2.72$ ^{7.05} |
| 14 30 | 33 $31.5 = 315 \div 10 \cdot = 3.5$ ^{7.05} | 52 Larger: dividing by a number less than 1 gives more (24). | |
| 15 5 | 34 $\times 100: 81 \div 3 \cdot = 27$ | | |
| 16 21 | 35 $4.8 \div 0.4 \cdot = 12 \text{ cups}$ | | |
| 17 8 | 36 $\times 100: 84 \div 3 \cdot = 28$ | | |
| 18 51 | 37 $1.6 \div 0.2 \cdot = 8 \text{ cups}$ | | |
| 19 15 | 38 7 right, 1 down ^{6.01} | | |

7.09 Fractions to decimals (booklet p. 19)

Examples (model these live)

E1 Write $\frac{3}{8}$ as a decimal.

$3.000 \div 8 \cdot = 0.375$

E2 Write $\frac{2}{3}$ as a decimal.

$2.000 \div 3 = 0.666... \cdot = 0.\dot{6}$

E3 Write $\frac{5}{11}$ as a decimal.

$5 \div 11 = 0.4545... \cdot = 0.4\dot{5}$

Questions (40 on today's skill, 17 from earlier lessons)

- | | | | |
|--------------------------------|---|--|--|
| 1 0.375 | 17 8.89 ^{7.07} | 33 $1 \div 6 = 0.16667... \cdot = 0.1\dot{6}$ | 46 $2 \div 7 = 0.285714286... \cdot 6 \text{ digits repeat}$ |
| 2 0.1 $\dot{6}$ | 18 $14 \div 25 \cdot = 0.56$ | 34 0.32 ^{7.06} | 47 $2.4 \div 0.4 \cdot = 6 \text{ cups}$ ^{7.08} |
| 3 $\frac{1}{3}$ | 19 $5 \div 8 \cdot = 0.625$ | 35 $\frac{3}{4} = 0.75 \cdot = 3.75$ | 48 $\frac{2}{5} = 0.4 \cdot = 1.4$ |
| 4 0.75 | 20 $1 \div 3 = 0.33333... \cdot = 0.\dot{3}$ | 36 $= 8 \frac{37}{100} \cdot = 8 \frac{37}{100}$ ^{7.02} | 49 $\frac{2}{5} = 0.4 \cdot = 2.4$ |
| 5 0. $\dot{6}$ | 21 0.800, 0.100, 0.579, 0.020 $\cdot$ 0.8, 0.579, 0.1, 0.02 ^{7.01} | 37 $1 \div 16 \cdot = 0.0625$ | 50 m^2 ^{5.03} |
| 6 $\frac{1}{9}$ | 22 $\frac{1}{8} = 0.125 \cdot = 2.125$ | 38 0.1 = 0.111... $\cdot$ larger: 0.1 | 51 $\frac{1}{4} = 0.25 \cdot = 3.25$ |
| 7 0.5 | 23 $1 \div 8 \cdot = 0.125$ | 39 0.3, 0.33, 0.375, 0.333... $\cdot$ 0.3, 0.33, $\frac{1}{3}$, $\frac{3}{8}$ | 52 14.4 ^{7.05} |
| 8 0.8 $\dot{3}$ | 24 $\times 100: 64 \div 8 \cdot = 8$ ^{7.08} | 40 $\frac{a}{4} \times 4 = 5 \times 4 \cdot a = 20$ ^{5.11} | 53 $\frac{5}{8} = 0.625 \cdot = 2.625$ |
| 9 0.08 ^{7.02} | 25 $5 \div 11 = 0.45455... \cdot = 0.4\dot{5}$ | 41 $1 \div 7 = 0.142857143... \cdot 6 \text{ digits repeat}$ | 54 $\frac{2}{5} = 0.4 \cdot = 4.4$ |
| 10 0.1 | 26 $\frac{2}{5} = 0.4 \cdot = 3.4$ | 42 0.3 ^{7.07} | 55 52.8 ^{7.06} |
| 11 $\frac{2}{3}$ | 27 $1 \div 11 = 0.09091... \cdot = 0.0\dot{9}$ | 43 $0.\dot{3} = 0.333... \cdot$ larger: 0.3 | 56 $\frac{1}{4} = 0.25 \cdot = 2.25$ |
| 12 0.8 | 28 $\frac{2}{5} = 0.4 \cdot = 5.4$ | 44 $3 \div 7 = 0.428571429... \cdot 6 \text{ digits repeat}$ | 57 $= \frac{45}{100} \cdot = \frac{9}{20}$ ^{7.02} |
| 13 equilateral ^{6.05} | 29 $a = 75$ ^{6.06} | 45 1 000 mL = 1 L $\cdot$ 345 $\div$ 1 000 = 0.345 L ^{7.04} | |
| 14 $7 \div 8 \cdot = 0.875$ | 30 $6 \div 25 \cdot = 0.24$ | | |
| 15 0.3 | 31 7.1 ^{7.03} | | |
| 16 0.95 | 32 0.6 = 0.666... $\cdot$ larger: 0.67 | | |

7.10 Rounding decimals (booklet p. 21)

Examples (model these live)

E1 Round 3.476 to 2 decimal places.
third place is 6: round up $\cdot \approx 3.48$

E2 Round 12.849 to 1 decimal place.
second place is 4: leave it $\cdot \approx 12.8$

E3 Round $\frac{5}{7}$ to 3 decimal places.
 $5 \div 7 = 0.71428\ldots \cdot \approx 0.714$

Questions (40 on today's skill, 16 from earlier lessons)

- | | | | |
|---|--|--|---|
| 1 34.6 | 16 5.72 | 30 look at place 2: $6 \cdot \approx 2.2$ | 43 $59 \div 3 = 19.6667\ldots \cdot \approx$
\$19.67 |
| 2 29 | 17 200 ^{3.01} | 31 0.62 ^{7.02} | 44 4.75 |
| 3 2.39 | 18 24.4 | 32 $75 \div 7 = 10.7143\ldots \cdot \approx$
\$10.71 | 45 $11.00 - 2.54 \cdot \approx 8.46$ ^{7.03} |
| 4 74.3 | 19 look at place 2: $7 \cdot \approx 8.3$ | 33 \$6.19 | 46 $58 \cdot 58.1, 58.09$ |
| 5 70 | 20 \$2.77 | 34 4.94 ^{7.07} | 47 7.95 |
| 6 2.14 | 21 0.330, 0.580, 0.500, 0.030
$\cdot 0.58, 0.5, 0.33, 0.03$ ^{7.01} | 35 $1 \div 6 = 0.1667\ldots \cdot \approx 0.17$ | 48 $27 \cdot 26.8, 26.80$ |
| 7 98.6 | 22 $2 \div 3 = 0.6667\ldots \cdot \approx 0.67$ | 36 $\frac{2}{5} = 0.4 \cdot \approx 3.4$ ^{7.09} | 49 56ac ^{5.08} |
| 8 25 | 23 look at place 4: $6 \cdot \approx 6.597$ | 37 look at place 2: $2 \cdot \approx 3.0$ | 50 $65 \div 6 = 10.8333\ldots \cdot \approx$
\$10.83 |
| 9 0.1 ^{7.09} | 24 $\times 100: 36 \div 9 \cdot \approx 4$ ^{7.08} | 38 $45 \div 7 = 6.4286\ldots \cdot \approx \6.43 | 51 0.0841 ^{7.04} |
| 10 12 | 25 \$9.40 | 39 6.75 | 52 $1 \div 7 = 0.1429\ldots \cdot \approx 0.14$ |
| 11 2.44 | 26 $5 \div 6 = 0.8333\ldots \cdot \approx 0.83$ | 40 $3 \text{ kg} = 3000 \text{ g} \cdot \frac{100}{3000} = \frac{1}{30}$
4.12 | 53 7.05 |
| 12 88.3 | 27 \$5.62 | 41 $65 \cdot 65.1, 65.14$ | 54 0.2 ^{7.07} |
| 13 52 cm ^{6.10} | 28 $5 \div 11 = 0.4545\ldots \cdot \approx 0.45$ | 42 7 000 ^{3.01} | 55 $33 \cdot 33.2, 33.25$ |
| 14 look at place 2: $1 \cdot \approx 1.0$ | 29 $a = 110$ ^{6.06} | | 56 $13 \div 40 \cdot \approx 0.325$ ^{7.09} |

7.11 Decimals and money (booklet p. 23)

Examples (model these live)

E1 Find the total: $\$4.85 + \$12.40 + \$0.95$
 $4.85 + 12.40 + 0.95 \cdot = \18.20

E2 Find the change from \$50 for \$37.65.
 $50.00 - 37.65 \cdot = \$12.35$

E3 Find the cost of 3.5 kg of apples at \$4.20 per kg.
 $3.5 \times 4.20 \cdot = \14.70

Questions (36 on today's skill, 15 from earlier lessons)

- | | | | |
|------------------------------------|---|--|---|
| 1 \$17.37 | 17 68 ^{7.10} | 29 $3 \times 3.46 \cdot = \$10.38$ | 40 4.13 ^{7.10} |
| 2 \$2.59 | 18 \$11.72 | 30 0.61 ^{7.02} | 41 $86.66 \div 5 = 17.3320\ldots \cdot \approx$
\$17.33 each; $5 \times 17.33 = \$86.65$ |
| 3 \$6.70 | 19 $4 \times 6.85 \cdot = \$27.40$ | 31 $23.18 \div 6 = 3.8633\ldots \cdot \approx$
\$3.86 each; $6 \times 3.86 = \$23.16$ | 42 total: \$12.69 $\cdot$ cash:
\$12.70 |
| 4 \$17.14 | 20 total: \$19.77 $\cdot$ change:
\$30.23 | 32 total: \$13.01 $\cdot$ change:
\$36.99 | 43 1 000 mL = 1 L $\cdot 877 \div 1$
000 = 0.877 L ^{7.04} |
| 5 \$6.88 | 21 0.050, 0.100, 0.624, 0.650
$\cdot 0.65, 0.624, 0.1, 0.05$ ^{7.01} | 33 4 ^{7.08} | 44 total: \$9.89 $\cdot$ cash: \$9.90 |
| 6 \$8.70 | 22 $4 \times 4.20 = 16.8 \cdot \approx \16.80 | 34 $2.6 \times 4.20 = 10.92 \cdot \approx$
\$10.92 | 45 $5.5 \times 4 = 22 \text{ hours} \cdot 22 \times$
24.50 = \$539.00 |
| 7 \$14.42 | 23 $9 \times 6.92 \cdot = \$62.28$ | 35 $50.00 - 12.72 \cdot = \$37.28$
7.03 | 46 5 right, 2 down ^{6.01} |
| 8 \$5.70 | 24 $4 \div 9 = 0.44444\ldots \cdot \approx 0.4$
7.09 | 36 $27.66 \div 6 = 4.6100\ldots \cdot \approx$
\$4.61 each; $6 \times 4.61 = \$27.66$ | 47 $20.72 \div 4 = 5.1800\ldots \cdot \approx$
\$5.18 each; $4 \times 5.18 = \$20.72$ |
| 9 2.7 ^{7.03} | 25 total: \$14.49 $\cdot$ change:
\$35.51 | 37 total: \$7.66 $\cdot$ cash: \$7.65 | 48 8.4 ^{7.05} |
| 10 \$4.10 | 26 total: \$25.02 $\cdot$ change:
\$24.98 | 38 $11p = 319 \cdot p = 29: 29$
pencils ^{5.13} | 49 $4 \times 1.20 = 4.8 \cdot \approx \4.80 |
| 11 \$8.85 | 27 $1.9 \times 1.20 = 2.28 \cdot \approx \2.28 | 39 $7.5 \times 5 = 37.5 \text{ hours} \cdot 37.5 \times$
26.50 = \$993.75 | 50 total: \$10.95 $\cdot$ cash:
\$10.95 |
| 12 \$13.89 | 28 $b - 12$ ^{5.04} | | 51 80 ^{7.08} |
| 13 $k = 30$ ^{5.10} | | | |
| 14 $9 \times 9.80 \cdot = \$88.20$ | | | |
| 15 \$16.43 | | | |
| 16 \$3.70 | | | |